

Diabetes Project Report

Predictive Modeling Using Machine Learning

1. Introduction

This project focuses on predictive modeling using machine learning techniques to predict whether a patient is diabetic or non-diabetic based on medical attributes. The diabetes dataset was analyzed, preprocessed, and used to train a Logistic Regression model.

2. Objectives

- Understand supervised learning concepts.
- Perform data preprocessing and cleaning.
- Train and test a machine learning model.
- Evaluate model performance using accuracy and confusion matrix.
- Visualize prediction performance.

3. Dataset Description

The dataset contains 768 records and 9 columns. Important features include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Sample Dataset Records

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
6.0	148.0	72.0	35.0	30.5	33.6	0.627	50.0	1.0
1.0	85.0	66.0	29.0	30.5	26.6	0.351	31.0	0.0
8.0	183.0	64.0	23.0	30.5	23.3	0.672	32.0	1.0
1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0

4. Data Preprocessing

Missing or invalid values represented by zeros in columns such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI were replaced with median values. The dataset was then split into training and testing sets. Feature scaling was applied using MinMaxScaler.

5. Machine Learning Model

Logistic Regression was used as the supervised learning algorithm for classification. The model was trained on normalized training data and tested using unseen test data.

6. Model Performance

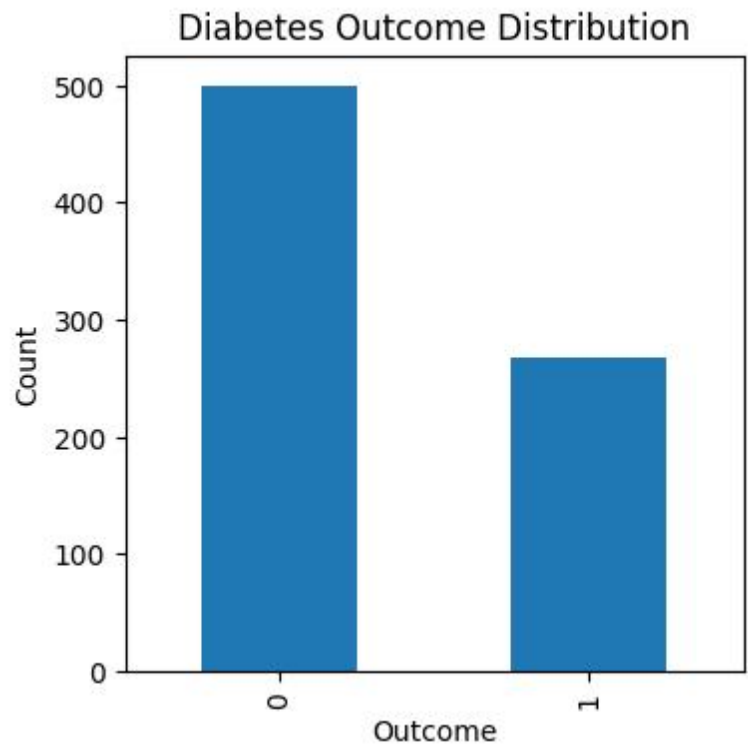
Model Accuracy: 0.74

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	102	21
Actual Diabetic	29	40

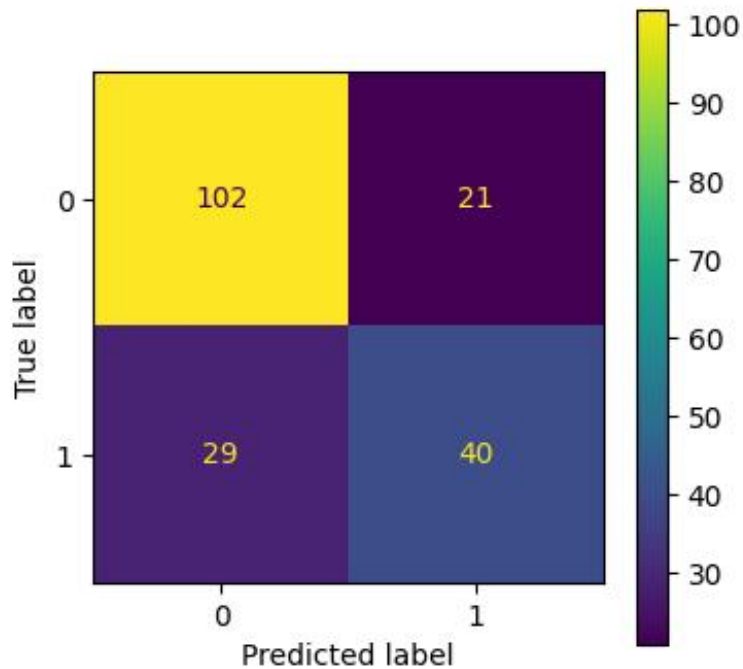
The confusion matrix shows that the model performed better in predicting non-diabetic patients compared to diabetic patients.

7. Visualizations

Outcome Distribution:



Confusion Matrix:



8. Conclusion

The project successfully demonstrated predictive modeling using machine learning. Logistic Regression achieved reasonable accuracy in predicting diabetes outcomes. This project provided practical experience in supervised learning, data preprocessing, model evaluation, and visualization.

9. Future Improvements

- Use advanced algorithms such as Random Forest and Decision Trees.
- Perform hyperparameter tuning for better accuracy.
- Apply cross-validation techniques.
- Deploy the model using a web application.